Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/04/2005
PI Name:	Howard, Heather Nicole		
Project Title:	Systematic Characterization of Antibiotic Effectiveness Under Altered Gravitational Environments (Simulated 0g to Centrifuged 10g)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80309	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	06/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Several experiments have indicated a trend toward decreased antibiotic effectiveness in vitro under spaceflight conditions. The microgravity environment encountered during spaceflight is thought to contribute to the observed changes. This proposed research will systematically characterize antibiotic effectiveness under altered gravitational environments (simulated 0g to centrifuged 10g). A thorough characterization of gravity dependent antibiotic resistance will provide a basis for future research into the mechanisms behind gravity dependent and/or spaceflight dependent alterations in antibiotic effectiveness.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	This research may help increase our understanding of multidrug resistance and improve our ability to treat bacterial infections.
Task Progress:	A methodology for investigating antibiotic effectiveness in clinostat generated simulated microgravity has been developed. Efforts are currently underway to validate the methodology against previous growth and antibiotic effectiveness studies in simulated microgravity. A comprehensive literature reveiw is also in progress.
Bibliography Type:	Description: (Last Updated:)